Load Priority Controller

Intelligent hardware power management for critical systems

"When power is scarce, priorities matter."

Group 2 - Tiny Tapeout

The Problem:

Power systems fail — and they fail fast.

Overloading a power supply causes instantaneous voltage collapse.

In embedded systems, solar setups, EVs, and off-grid nodes, this kills the entire system — including critical subsystems that didn't even need to go offline.

"A 100ms power fault is enough to crash an embedded controller."

Microcontrollers can’t react fast enough.

- Firmware-based solutions introduce boot latency, interrupt jitter, and software bugs.
- A MCU under power stress may itself crash before it can shed any loads.
- Firmware needs maintenance; hardware logic does not.

"Hardware reacts in nanoseconds. Firmware reacts in milliseconds — or not at all."

Feature	Software	Hardware FSM
Speed	Milliseconds	**Nanoseconds**
Reliability	Jittery/Bugs	**Deterministic**
Complexity	High (FW)	Low (Logic)
Failure	System Crash	**Safe State**

A pure-hardware solution. No CPU. No firmware. No failure.

The Load Priority Controller is a Finite State Machine implemented entirely in digital logic. It reads voltage-level signals in real time and decides which loads stay on automatically, every clock cycle.

LPC

3 Power States

4 Managed Loads

1 Cycle Response

0 Lines Firmware

TECHNICAL OVERVIEW

How It Works: From voltage sensing to load control in one pipeline.

Battery / Power Bus → Voltage Comparator (high_power, medium_power, undervoltage) → **LPC FSM** → 4 Load Enable Lines → MOSFETs / Relays → Loads L1–L4

*The comparator is the **sensor**. The LPC is the **brain**. The MOSFETs are the **muscle**.*

Three modes. One rule: *Protect what matters most.*

LOW POWER MODE
Only L1 ON

0001

Triggered by undervoltage

Critical systems only

MEDIUM POWER MODE
L1 + L2 ON

0011

Triggered by medium flag

Core operations

HIGH POWER MODE
All 4 Loads ON

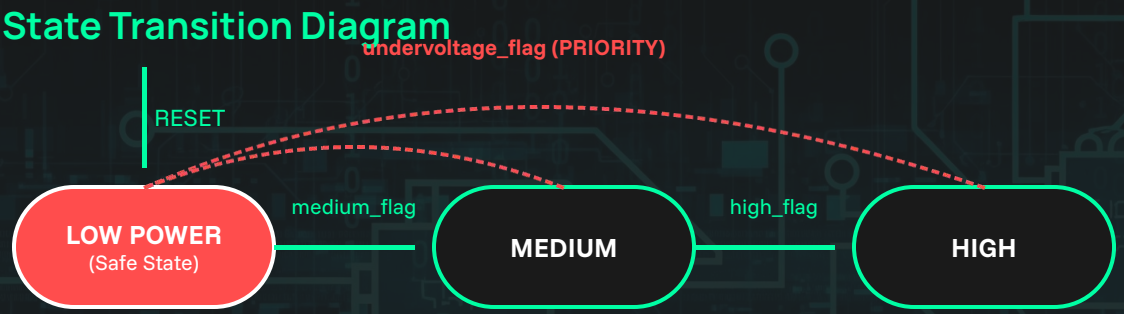
1111

Triggered by high flag

Full capacity

A 3-state Finite State Machine with built-in priority logic.

State Transition Diagram



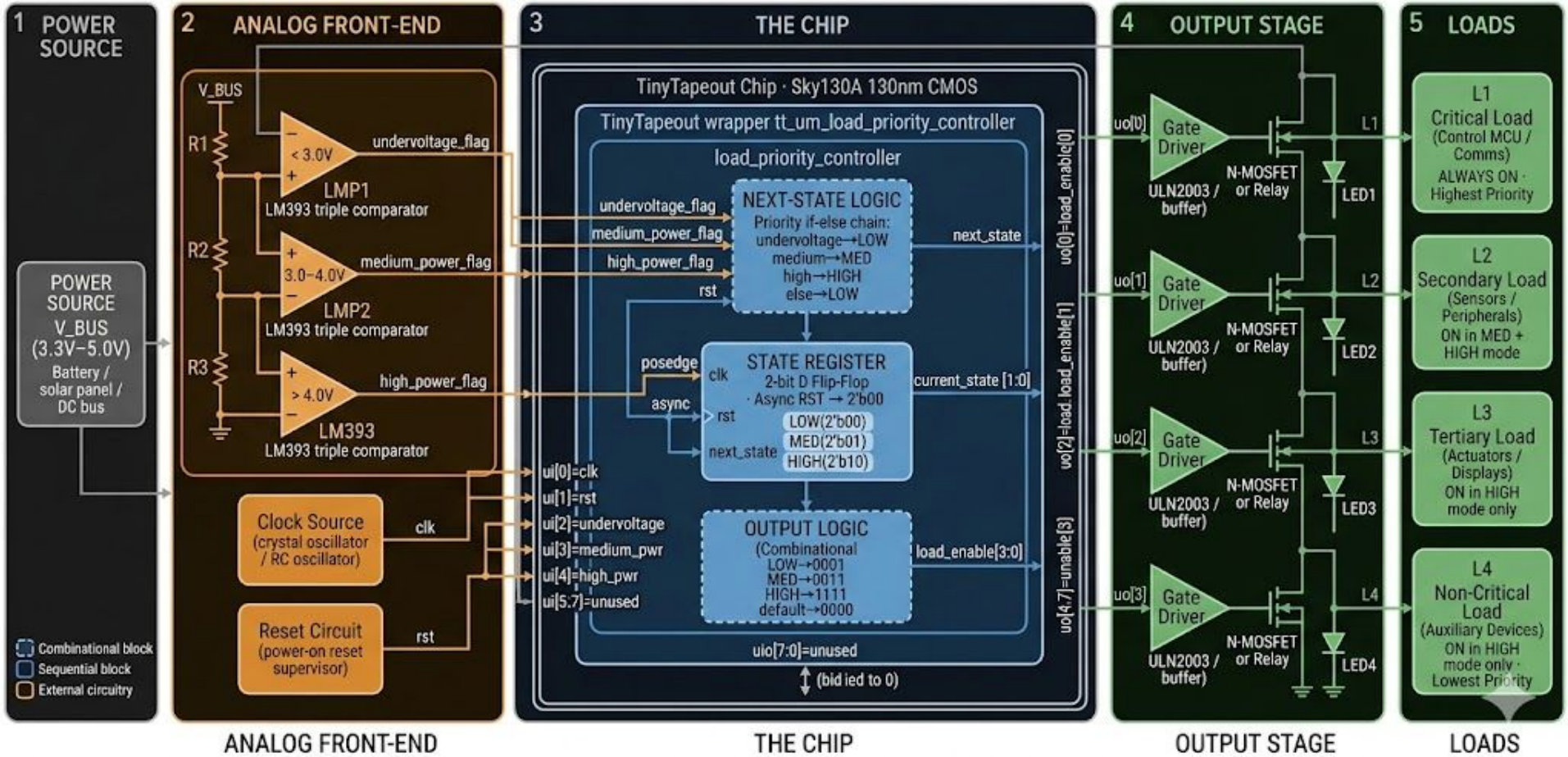

```
// Priority Logic
if (undervoltage_flag)
  next_state = LOW_POWER;
else if (high_power_flag)
  next_state = HIGH;
else if (medium_power_flag)
  next_state = MEDIUM;
```

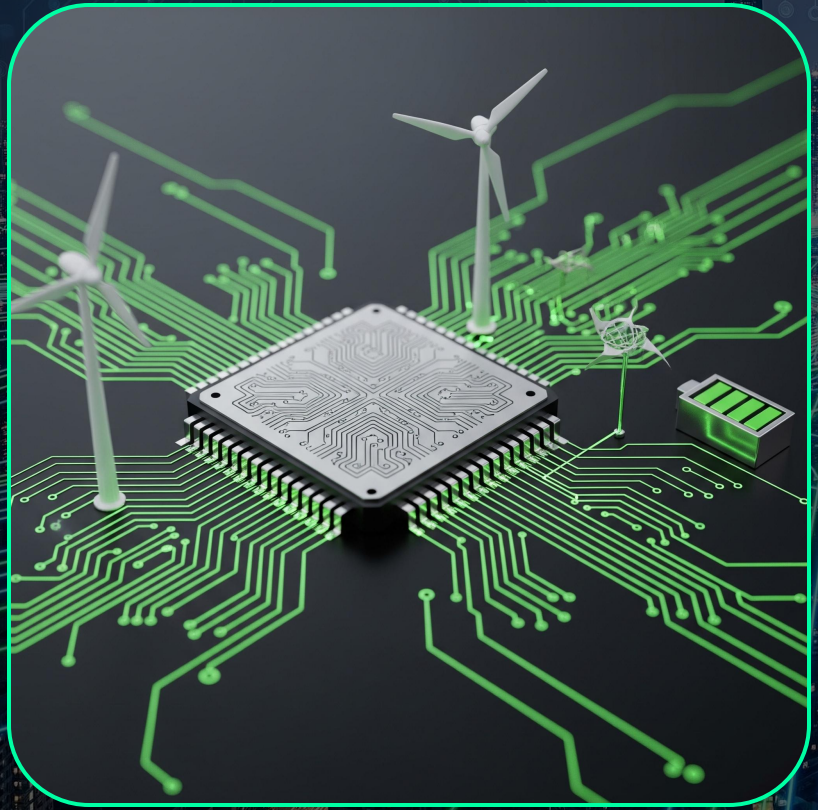
undervoltage_flag always wins

Safety is non-negotiable.

Load Priority Controller — Complete System Block Diagram

Sky130A 130nm CMOS · TinyTapeout · 3-State FSM · 4 Prioritized Load Outputs





Applications

A REAL WORLD SOLUTION.

Efficiently managing power across diverse industries:

- Solar/Battery Microgrids
- Electric Vehicle Systems
- Off-Grid IoT Infrastructure
- Industrial Automation & Safety

Built for any system where power is a constraint.

Real-World Use Cases

Built for any system where power is a constraint.

Solar/Battery Microgrids
Shed non-critical loads when solar input drops.

Electric Vehicles
Protect drivetrain systems during battery sag.

Off-Grid IoT Nodes
Keep comms alive; cut sensors when energy is low.

Hospital Backup
ICU stays on; non-critical rooms shed load.

Spacecraft/Satellites
Fault-tolerant, no firmware to corrupt.

Industrial Automation
Protect PLCs and safety systems first.

Why This Matters: Key Advantages

Hardware wins where software cannot.

Speed

Nanosecond response vs. millisecond firmware latency.

Reliability

No OS, no stack overflow, no boot sequence.

Scalability

Add more comparator thresholds, more load lines.

Silicon-Proven

Designed for real ASIC fabrication on Sky130A.

Verification and Testing

Tested in simulation. Validated in waveforms.

RTL simulation with Icarus Verilog

Python-driven test cases using cocotb framework

Waveform analysis in GTKWave

Automated CI/CD via GitHub Actions — every commit triggers a full test + GDS build

Scenario	Input Flags	Expected
Reset	RST=1	L0-L3: OFF
Normal	V>TH1	All loads ON
Critical	V<TH2	L0 ON, L1-3 OFF
Shedding	V<TH1	L0-1 ON, L2-3 OFF

Conclusion & What's Next

A hardware-first approach to power intelligence.

- **Pure FSM logic** — no firmware dependency
- **Priority-driven load shedding** in one clock cycle
- **Real ASIC submission** on Sky130A
- **Extensible** to more states, more loads, adaptive thresholds

"Power management shouldn't rely on software that can crash. We built the alternative."

The Team

Designed with purpose. Built for silicon.

Abiola Enoch
Chip Designer 1
gITHUB

Omotosho Enoch
Team Lead
GITHUB

Esabu Blessing
Chip Designer 2
GITHUB

THANK YOU!

View on GitHub: https://github.com/R0B0T003/Load-Priority-Controller-TinyTapeout.git